Iknoor Braich

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Education

University of Guelph **Sep 2024 – Present**
Bachelor of Engineering – Mechanical Engineering (Co-op) *Guelph,Ontario,Canada*

- **Overall GPA:** 3.7 - Dean's Honours List 2025 – Present
- **Certifications:** CSWA, MATLAB/Simulink Onramp

Experience

Mechanical Design Member (Brakes) **September 2024 – Present**
Gryphon Racing Formula SAE *Guelph, Ontario Canada*

- Mitigated competition risk under a 1-week deadline ahead of FSAE Michigan 2026 by reverse engineering a 32mm throttle body in SW from a purchased unit and adapting it for 3D-printed production, delivering a compliant mechanical backup after the electronic throttle body remained unapproved until just before competition.
- Reduced pedal box mass by 15% while maintaining strength and safety margins, by performing FEA to identify low-stress regions and redesigning accordingly.
- Supported manufacturing of 4 brake system components by producing detailed engineering drawings, including the pedal box, brake inserts, and accelerator pedals with complete dimensioning, GD&T tolerancing, and fabrication notes.
- Documented the complete brake line routing architecture against the car's chassis in AutoCAD, creating a reference diagram to improve new members' understanding of the system layout.
- Prepared a composite nose cone for carbon fiber layup by hand-sanding the surface to the required finish while wearing full PPE, following standard composite prep procedure ahead of layup.

Projects

Autonomous Payload Drone | *Fusion360, Ansys Fluent, FDM 3D-Printing* **May 2026 - September 2026**

- Worked with a team of 2, led the Mechanical Design of a complete custom drone frame and mounting architecture from scratch to house all critical electronics, including ESC, battery, motor mounts, and a dedicated Pixhawk 6C flight controller mount positioned with 10mm clearance above the ESC; done by iterating across 3 prototype designs before finalizing geometry for FDM production.
- Reduced overall drone weight by 35% (1.5kg total) while increasing frame bending strength by transitioning material selection from PETG to PA6-CF, based on comparative analysis of mechanical properties using Bambu Lab's material data sheets.
- Achieved secure, tool friendly assembly by designing heat-set insert extrusions (M3 & M5) for structural fastening and snug-fit clamps for carbon fiber tube profiles, refined through iterative prototyping to ensure a precise mechanical fit.
- Achieved autonomous stage 1 flight, confirming successful integration of frame, propulsion, and control systems.

Rotary Inverted Pendulum | *C++, Fusion360* **June 2026 – July 2026**

- Adapted a custom PID control system to a novel hardware configuration, achieving accurate real-time pendulum motion detection and correction response, by modifying an open-source algorithm's control logic and manually tuning gains to fit this project's specific mechanical parameters.
- Isolated a hardware-level performance limitation from the control software by systematically testing PID behavior under load, confirming intermittent motor slip traced back to thermal deformation in the 3D-printed stepper mount rather than a flaw in the control logic itself.
- Enabled real time closed loop position feedback by integrating an AS5600 magnetic encoder with an Arduino via I2C Fast Mode (400kHz), reading pendulum angle to drive PID correction at high update rates.
- Delivered a fully functional control system for under $20 CAD by designing a custom 3D-printed housing to mount the Arduino, NEMA 17 stepper motor, and encoder as a single integrated assembly.

Kinetic Energy Storage & Conversion Toy - Eng Design 2 | ***SolidWorks*** **September 2025 – December 2025**

- Engineered a spring driven mechanism to store potential energy and convert it into sustained kinetic motion via a rack & pinion gear train, designing the assembly to collapse within a Kinder egg's packaged profile and expand into a larger, functional device; validated across 3 SolidWorks design iterations
- Achieved reliable 2m travel from a single spring load by systematically testing and tuning spring stiffness, rack travel distance, and gear ratios to meet both the motion target and the egg's size constraint.
- Created a BOM to track project expenses and assessed feasibility for theoretical mass production by conducting a market analysis evaluating competitor pricing, local injection molding options and the carbon impact of material choices.

Technical Skills

Design/Analysis: Solidworks, Fusion 360, Siemens NX, AutoCAD, Ansys Fluent, Prototyping, DFMA, GD-T
Software: C++, Python, MATLAB
Manufacturing: 3D Printing, Fabrication (Mill, Lathe, Drill Press), Multimeter, Soldering, Welding